

# Max Zhu

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## OBJECTIVE

Motivated Computer Engineering student strongly interested in software engineering, full-stack development, and embedded systems. Seeking a Summer 2025 internship to gain real world experience in high- and low-level programming, digital design, hardware-software integration, and machine learning. Eager to contribute to innovative projects, determined to expand my knowledge, and excited to collaborate with a like-minded team to solve complex challenges in a dynamic environment.

## EDUCATION

### GEORGIA INSTITUTE OF TECHNOLOGY

#### B.S. in Computer Engineering

- **Threads:** Systems Architecture and Computing Hardware and Emerging Architecture

Atlanta, GA

Expected Graduation *May 2026*

**GPA: 3.9/4.0**

## EXPERIENCE

### Georgia Tech School of Electrical and Computer Engineering

#### Opportunity Research Scholar for Wireless Twin Digital Systems

- Analyzed and annotated 50+ research papers on 5G, Wi-Fi 6, ray tracing, wireless signals, and UAVs
- Utilized a combination of Sionna, Blender, Wireless Insight, and radio environments create of wireless twin digital system

Atlanta, GA

*August 2024 – Present*

#### Knack

##### Student Tutor

- Created tailored learning plans for 30+ students to teach complex concepts and maintain engagement
- Taught courses Digital System Design, Circuit Analysis, Data Structure and Algorithms, and Object-Oriented Programming

Atlanta, GA

*August 2024 – Present*

### Metro Healthcare

#### IT Specialist

- Integrated digital infrastructure like POS systems, shared drives, website access, and patient database across multiple locations
- Developed a clinical website to showcase services and streamline appointment scheduling

Vancouver, BC

*November 2023 – January 2024*

## PROJECTS

### Personal Website - maxzhu143.github.io

- Designed and developed an engaging website using HTML, CSS, and JavaScript to showcase projects

*August 2024 - Present*

### Cloud Server

- Configured and deployed a cloud server with Ubuntu Server to replace multiple personal subscription services
- Exclusively utilized Linux CLI (command-line interface) for system administration and server organization

*July 2024 – Present*

### Spotify Car Thing

- Developed a desktop device inspired by the “Spotify Car Thing” allowing for live interactions with the Spotify interface
- Integrated physical inputs and utilized the ESP8266, C++, and Spotify API to skip, like, pause, and view the current

*September 2024 – November 2024*

### Atlanta Food Finder

- Developed a Django-based web application that allows users to discover nearby personalized restaurant recommendations
- Worked as a full-stack developer, using Google API, Python, CSS, and HTML to host the website with Heroku

*August 2024 – October 2024*

## LEADERSHIP AND EXTRACURRICULARS

### SILICON JACKETS

#### Digital Design Member

- Designed and implemented an arithmetic logic unit on a CPU using Verilog, focusing on state machines for optimized performance
- Developed and executed comprehensive test benches to verify hardware designs, ensuring accuracy and reliability

Atlanta, GA

*August 2024 – Present*

### HIVE

#### Peer Instructor

- Taught and guided end-users on laser cutters, 3d printers, shop machines, and benchtop tools
- Led interactive workshops on circuit design, skill implementation, and product development

Atlanta, GA

*December 2023 – Present*

## SKILLS

**Software:** C, C++, VHDL, Java, Python, Verilog, Git, CUDA, Ubuntu, MatLab, Sionna, Quartus, HTML, Javascript

**Hardware:** FPGA, Oscilloscopes, Digital Circuits, KiCad, Arduino, CAD, Software Defined Radio, Signal Analyzer